

Introduction to Social Influence Models

Auto-Logistic Actor Attribute Models (ALAAMs)

Johan Koskinen

Department of Statistics
Stockholm University
University of Melbourne

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Stockholm
University

Preamble

- All material is on the workshop repository
<https://github.com/johankoskinen/ALAAM>
 - ▶ Download the RMarkdown file CHDH-SNA-1.Rmd
 - ▶ Download the (proto) manual https://github.com/johankoskinen/ALAAM/blob/main/alaam_effects.pdf
 - ▶ This afternoon CHDH-SNA-2025-2.Rmd and a selection of other examples
- In order to run the Markdown you need
 - ▶ The R-package 
 - ▶ The RStudio interface  RStudio
- We will predominantly use the packages
 - ▶ sna
 - ▶ network
- as well as **ba1aam.R** from GitHub



What is new

If you have used BayesALAAM before

- Entirely new
 - ▶ MultivarALAAM.R $\Rightarrow$ balaam.R
 - ▶ Documentation: `alaam.effects.pdf`
- Define and estimate the model using
 - ▶ Standard formula agree $\sim$ odegree + mood + sex + simple
 - ▶ Main function `estimate.alaam` returns `estimate.alaam.obj`
- The object `prevBayes`
 - ▶ Continue previous estimation `estimate.alaam.obj`
 - ▶ recalibrate the proposal variance-covariance matrix
- Model selection
 - ▶ Obtain posterior deviance from `post.deviance.alaam` applied on `estimate.alaam.obj`
 - ▶ Calculate DIC using `alaam.dic` directly on object returned by `post.deviance.alaam`
- ... and a lot of other tweaks that may or may not have broken the functionality



Agenda

1

Preamble

- Data
- Homophily
- Autocorrelation

2

ALAAM

- Markov
- Contagion
- Effects

3

Estimation

- Monitoring performance

4

GOF

5

Further topics



Data structure and homophily

Data structure and homophily

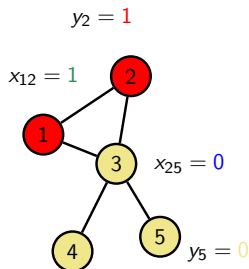


Data - binary outcomes

Tie-variables:

$$X_{ij} = \begin{cases} 1, & \text{if tie from } i \text{ to } j \\ 0, & \text{else} \end{cases}$$

Adjacency matrix



$$\mathbf{X} = (X_{ij})_{ij \in V \times V} = \begin{bmatrix} \cdot & \text{green } 1 & 1 & 0 & 0 \\ \text{green } 1 & \cdot & 1 & 0 & \text{blue } 0 \\ 1 & 1 & \cdot & 1 & 1 \\ 0 & 0 & 1 & \cdot & 0 \\ 0 & \text{blue } 0 & 1 & 0 & \cdot \end{bmatrix}$$

Nodes: $V = \{1, 2, \dots, n\}$ **Attribute vector**

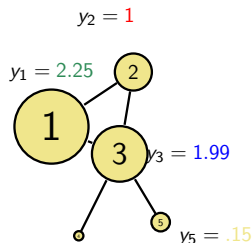
$$\mathbf{y} = [\text{red } 1 \quad \text{red } 1 \quad \text{yellow } 0 \quad \text{yellow } 0 \quad \text{yellow } 0]^T$$



Data - Continuous outcomes

Tie-variables:

$$X_{ij} = \begin{cases} 1, & \text{if tie from } i \text{ to } j \\ 0, & \text{else} \end{cases}$$



Adjacency matrix

$$\mathbf{X} = \begin{bmatrix} \cdot & \textcolor{green}{1} & 1 & 0 & 0 \\ \textcolor{green}{1} & \cdot & 1 & 0 & \textcolor{blue}{0} \\ 1 & 1 & \cdot & 1 & 1 \\ 0 & 0 & 1 & \cdot & 0 \\ 0 & \textcolor{blue}{0} & 1 & 0 & \cdot \end{bmatrix}$$

Nodes: $V = \{1, 2, \dots, n\}$

Attribute vector

$$\mathbf{y} = [\textcolor{green}{2.25} \quad \textcolor{red}{1} \quad \textcolor{blue}{1.99} \quad \textcolor{yellow}{.05} \quad \textcolor{yellow}{.15}]^T$$

Homophily

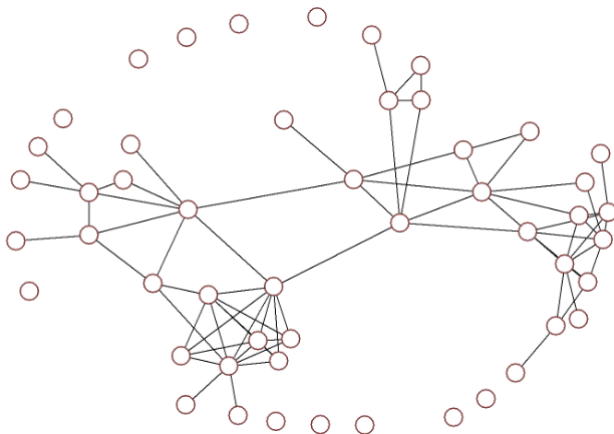


Homophily - empirical evidence

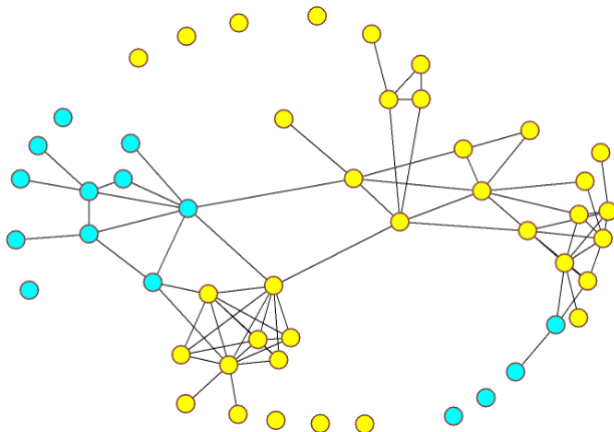
We repeatedly **observe** that people that are *similar* hang together



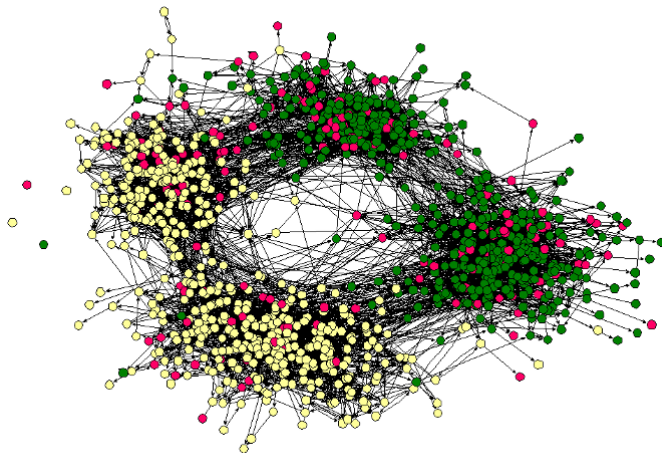
Yuval Kalish - school kids Israel (1)



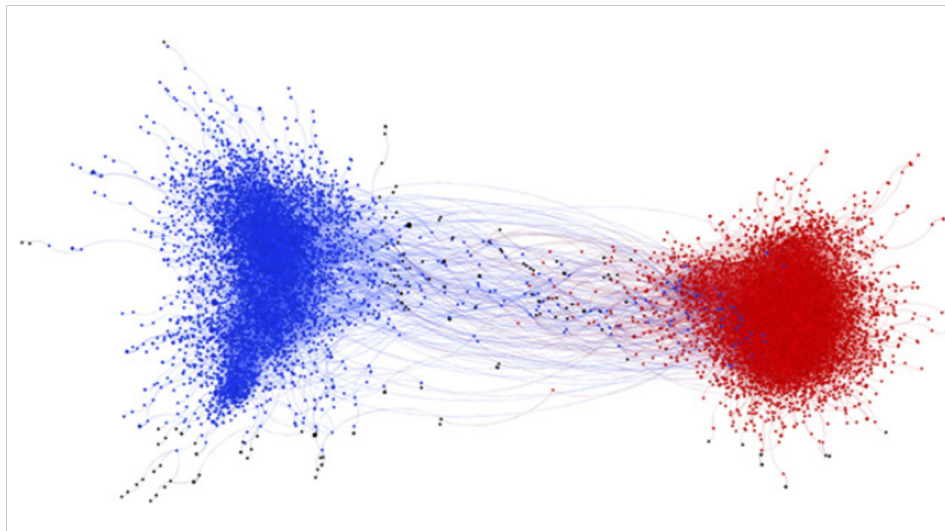
Yuval Kalish - school kids Israel (2)



Ad Health - school kids of different races (Moody et al)



Democrats and Republicans on twitter (Brady et al., 2017)

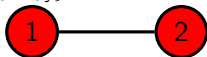


Homophily

Hanging together, being similar

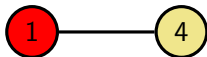
Binary Y_i :

$$\mathbb{1}_{\{y_i=y_j\}}x_{ij}$$



$$y_1 = 1$$

$$y_2 = 1$$

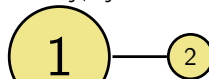


$$y_1 = 1$$

$$y_2 = 0$$

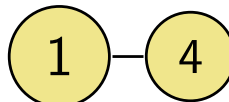
Continuous Y_i :

$$|y_i - y_j|x_{ij}$$



$$y_1 = 2.25$$

$$y_2 = 1$$



$$y_1 = 2.25$$

$$y_2 = 1.99$$

Example data

Example (S50 Data)

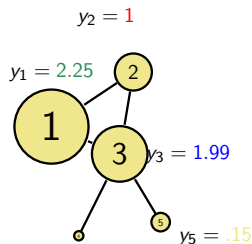
Look at a dataset of 50 Glaswegian school girls

https://www.stats.ox.ac.uk/~snijders/siena/s50_data.htm

In CHDH-SNA-1.Rmd:

```
X <- as.matrix( read.table( "s50-network1.dat" ) )
sport <- read.table( "s50-sport.dat" )
smoke <- read.table( "s50-smoke.dat" )
alcohol <- read.table( "s50-alcohol.dat" )
n <- nrow(X)
smoke[smoke[,2]==1 ,2] <- 0
smoke[smoke[,2]==2 | smoke[,2]==3 ,2] <- 1
sport <- sport[,1]-1
gplot( X )
```

Correlations of continuous outcomes (1)



Nodes: $V = \{1, 2, \dots, n\}$

Correlation for variables U and V :

$$\text{corr}(U, V) = \frac{\sum_{i=1}^n (u_i - \bar{u})(v_i - \bar{v})}{\sqrt{\sum_{i=1}^n (u_i - \bar{u})^2 \sum_{i=1}^n (v_i - \bar{v})^2}}$$

For **network correlation**
we only need associations

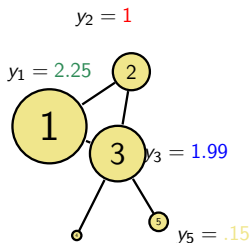
$$(y_i - \bar{y})(y_j - \bar{y})$$

Between i and j that are connected, i.e.

$$x_{ij} = 1$$

Correlations of continuous outcomes (2)

Moran's I:



Nodes: $V = \{1, 2, \dots, n\}$

$$I_k = \frac{n \sum_{i=1}^n \sum_{j=1}^n (y_i - \bar{y})(y_j - \bar{y}) x_{ij}^{(k)}}{\sum_{i,j} x_{ij}^{(k)} \sum_{j=1}^n y_j^2}$$

Where

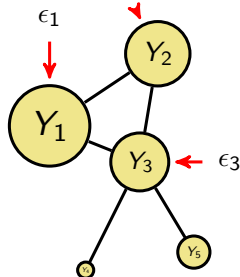
$$x_{ij}^{(1)} = x_{ij}$$

and $x_{ij}^{(2)}$ if i and j are at a distance of 2, etc
If $l_1, l_2, \dots$ are large, neighbouring nodes have similar values

Network autocorrelation model

The **Network autocorrelation model**:

$$\epsilon_2 = \rho\epsilon_1 + \rho\epsilon_3 + \xi_2$$



Nodes: $V = \{1, 2, \dots, n\}$

$$Y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik} + \epsilon_i$$

assumes a linear model where there is
autocorrelation between the errors

$$\epsilon_i = \rho \sum_{j=1}^n w_{ij} \epsilon_j + \xi_i, \quad \xi_i \stackrel{iid}{\sim} N(0, \sigma^2)$$

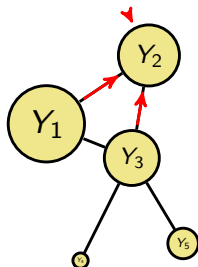
where w_{ij} is either x_{ij} or a scaled version of it

Note: if $\rho = 0$, we have regular regression with independent observations

Network effects model

The **Network effects model**:

$$Y_2 = \rho Y_1 + \rho Y_3 + \epsilon_2$$



Nodes: $V = \{1, 2, \dots, n\}$

$$Y_i = \rho \sum_{j=1}^n w_{ij} Y_j + \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik} + \epsilon_i$$

where w_{ij} is either x_{ij} or a scaled version of it, assumes a non-linear model where there is **autocorrelation between the outcomes**. Errors are assumed independent

$$\epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$$

Note: if $\rho = 0$, we have regular regression with independent observations

ALAAM - The basic model

The Model



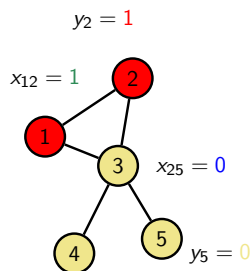
Data - binary outcomes

We want to model binary **Attribute vector**

$$\mathbf{y} = [\textcolor{red}{1} \ \textcolor{red}{1} \ \textcolor{yellow}{0} \ \textcolor{yellow}{0} \ \textcolor{yellow}{0}]^T$$

conditional on **Adjacency matrix**

$$\mathbf{X} = \begin{bmatrix} \cdot & \textcolor{green}{1} & 1 & 0 & 0 \\ \textcolor{green}{1} & \cdot & 1 & 0 & \textcolor{blue}{0} \\ 1 & 1 & \cdot & 1 & 1 \\ 0 & 0 & 1 & \cdot & 0 \\ 0 & \textcolor{blue}{0} & 1 & 0 & \cdot \end{bmatrix}$$



Marginally, we can think of modelling the probabilities

$$p_i = \mathbb{E}(Y_i \mid \mathbf{X}) = \Pr(Y_i = 1 \mid \mathbf{X})$$

Logit: Log-odds

Odds

For a probability p , we define the odds as

$$\frac{p}{1-p}.$$

which is always positive, and increases with p (e.g. $\frac{0.9}{0.1} > \frac{0.5}{0.5} > \frac{0.1}{0.9} > 0$)

Logit

The logarithm of the odds, the log odds, is called *logit*

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right)$$

which increases with p but $\text{logit}(p)$ takes all values in $\mathbb{R}$

The logit maps probabilities to all values in $\mathbb{R}$

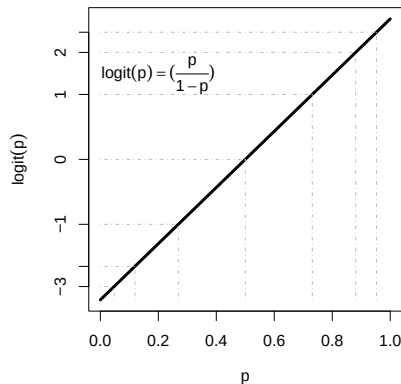
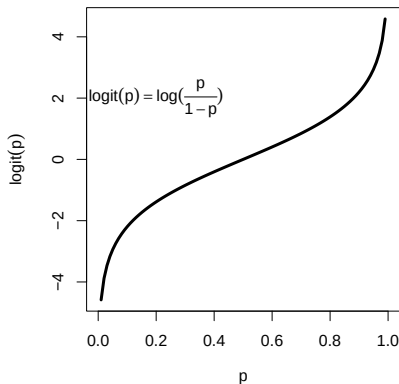


Figure: Probabilities against logit function. Logit scale on vertical axis (right)

Logit link function and linear predictors

While

$$0 \leq p \leq 1$$

as $\text{logit}(p) \in \mathbb{R}$, we write

$$\text{logit}(p_i) = \eta_i$$

where η_i is the **linear predictor**

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}$$

which is a linear regression



The logit-link function

For any $x_{i1}, \dots, x_{ik}$, and parameters $\beta_0, \beta_1, \dots, \beta_k$, we can calculate

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik}$$

and use the *inverse logit*

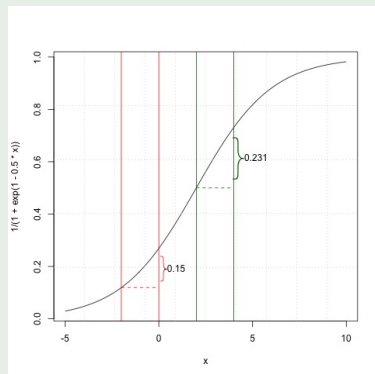
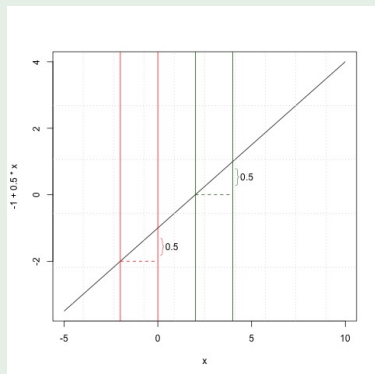
$$\underbrace{\eta_i = \log \left(\frac{p_i}{1 - p_i} \right)}_{\text{logit}} \quad \underbrace{\Rightarrow}_{\text{solve for } p_i} \quad p_i = \underbrace{\frac{e^{\eta_i}}{1 + e^{\eta_i}}}_{\text{inverse logit}}$$

to calculate the probability p_i

Probabilities and the linear predictor: non-linearity

Example (Hypothetical example)

Assume $\text{logit}[\mathbb{E}(Y \mid x)] = \eta$, where $\eta = -1 + 0.5x$



$$\eta_1 - \eta_0 = \beta_1(x_1 - x_0), \text{ but } \text{logit}^{-1}(\eta_1) - \text{logit}^{-1}(\eta_0)$$

Binary outcomes

We want to model

$$p_i = \Pr(Y_i = 1 \mid \mathbf{X})$$

and if the Y_i are independent

$$p(\mathbf{y} \mid \mathbf{X}) = \prod_{i=1}^n \Pr(Y_i = y_i \mid \mathbf{X})$$

where for each $i = 1, \dots, n$

$$\Pr(Y_i = 1 \mid \mathbf{X}) = \frac{e^{\eta_i}}{1 + e^{\eta_i}}$$



Auto-Logistic Actor Attribute Model (ALAAM)

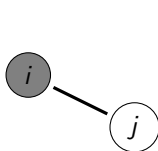
What if we let $\Pr(Y_i = 1|\mathbf{X})$ depend on i 's position in the network?
For example

$$\eta_i = \beta_0 + \beta_{\text{deg}} \sum_j x_{ij} + \beta_{\text{var}} \sum_{j,k} x_{ij} x_{ik} + \beta_{\text{tri}} \sum_{j,k} x_{ij} x_{ik} x_{jk}$$

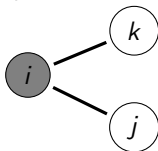
which gives us a model

$$p(\mathbf{y} | \mathbf{X}) = \exp \left\{ \boldsymbol{\beta}^\top \mathbf{z}(\mathbf{y}, \mathbf{X}) - \psi(\boldsymbol{\beta}) \right\}$$

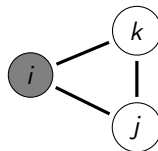
where $\mathbf{z}(\mathbf{y}, \mathbf{X}) = (z_1, \dots, z_p)^\top$, $z_1 = \sum y_i$, and



$$z_2 = \sum_i y_i x_{i+}$$



$$z_3 = \sum_i y_i \sum_{j,k} x_{ij} x_{ik}$$



$$z_4 = \sum_i y_i \sum_{j,k} x_{ij} x_{ik} x_{jk}$$

Auto-Logistic Actor Attribute Model (ALAAM)

If $\beta_{\text{deg}} > 0$ then nodes with high degree centrality are more likely to have $y_i = 1$ than nodes with low degree



The network activity ALAAM - Markov model

Frank and Strauss (1986) derived a model for interdependent network *ties* from a Markov dependence assumption. For **attributes**:

Markov dependence assumption (Robins et al., 2001)

Considering the collection of variables $\mathbf{M} = (\mathbf{y}, \mathbf{X})$ Let variables M_u and M_v be conditionally independent if $u \cap v = \emptyset$

Example (Conditionally dependent variables)

The outcomes Y_i and X_{ij} are conditionally dependent as $\{i\} \cap \{i, j\} = \{i\}$

Example (Conditionally independent variables)

The outcomes Y_i and X_{kj} are conditionally independent as $\{i\} \cap \{i, j\} = \emptyset$



Deriving model from dependence (as in ERGM)

Network Block Attribute Block

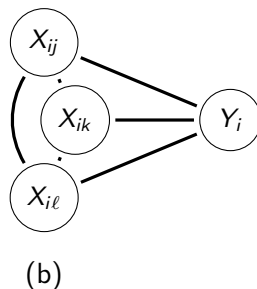
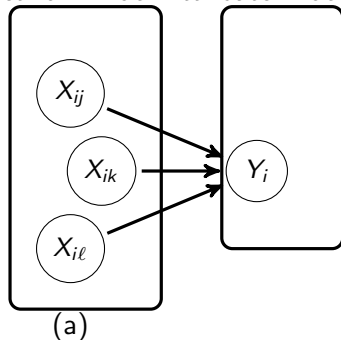


Figure: Dependence graph (a) and Moral graph (b) of network activity dependence model (Robins et al., 2001)

The network activity ALAAM - Markov model

The statistics z_r correspond to cliques in the Moral graph, and includes

- intercept: $\sum y_i$
- degree: $\sum y_i \sum_j x_{ij}$
- stars: $\sum y_i \sum x_{ij_1} \cdots x_{ij_k}$

But crucially, **no** statistics of the type

$$y_i y_j x_{ij}$$

and thus Y_i and Y_j are independent given $\mathbf{X}$

$$\Pr(Y_i = y_i, Y_j = y_j \mid \mathbf{X}, \mathbf{y}_{-i,j}) = \Pr(Y_i = y_i \mid \mathbf{X}, \mathbf{y}_{-i,j}) \Pr(Y_j = y_j \mid \mathbf{X}, \mathbf{y}_{-i,j})$$



The network activity ALAAM - logistic regression

The network activity ALAAM is equivalent to logistic regression with

$$\text{logit}(p_i) = \beta_0 + \beta_1 z_{i1} + \cdots + \beta_p z_{ip}$$

where the statistics z_{ih} are summaries of i 's network position



The network activity ALAAM - logistic regression

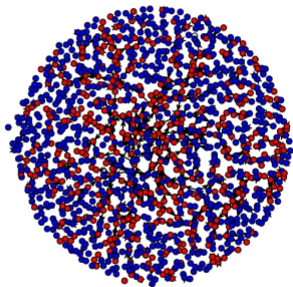
Example (Modern contraceptive use in rural Kenya)

	Mean	Description
mcUse	0.35	Do you use modern contraceptive techniques?
Age	34.41	Age (sd:16.04)
Female	0.60	Female (1) or Male (0)
HasChildren	0.68	Have one child or more
relevanOthersApprove	0.45	Other people's approval is important
relevanOthersUse	0.67	I care if other people use modern contraceptives
mcUseConflict	0.68	The use of modern contraceptives is contentious and causes conflict
numFriends	0.88	Tallied: the number of names of people they spend their free time with

Table: Variables in Kenya study on Modern contraception usage (Not exact question wordings)(NSF-CMMI-2005661)Modi, Koskinen, DeChurch, Contractor, 2025, SocNet

Data - dichotomous outcome conditional on network

Example (Modern contraceptive use in rural Kenya (cont.) $n = 1303$)



X: Spending free time

y: Use modern contraceptives

The network activity ALAAM - logistic regression

Example (Modern contraceptive use in rural Kenya (cont.) $n = 1303$)

Estimated logistic regression

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.6340	0.2601	-2.44	0.0148
Age	-0.0554	0.0067	-8.24	0.0000
Female	-1.0232	0.1538	-6.65	0.0000
HasChildren	1.9622	0.2068	9.49	0.0000
relevantOthersApprove	1.4696	0.1514	9.70	0.0000
relevantOthersUse	0.3415	0.1720	1.99	0.0471
mcUseConflict	-0.3835	0.1474	-2.60	0.0093
numFriends	0.3349	0.0828	4.04	0.0001

How much is the increase in the probability of mcUse if you acquire another friend?

Example data

Example (Modern contraceptive use in rural Kenya (cont.)) $n = 1303$

Using balaam

```
res.0 <- estimate.alaam( mcUse ~odegree
                        + Age
                        + Female
                        + HasChildren
                        + relevanOthersApprove
                        + relevanOthersUse
                        + mcUseConflict,
                        village , adjacency = adj,
                        Iterations = 2000)
```

Example data

Example (Modern contraceptive use in rural Kenya (cont.) $n = 1303$)

	Posterior		95% CI	
	Estimate	sd	0.025	0.975
intercept	-0.593	0.253	-1.102	-0.081
contagion	0	0	0	0
odegree	0.337	0.083	0.16	0.518
Age	-0.057	0.006	-0.071	-0.045
Female	-1.061	0.15	-1.345	-0.753
HasChildren	2.014	0.243	1.555	2.489
relevanOthersApprove	1.479	0.146	1.18	1.752
relevanOthersUse	0.335	0.168	0.005	0.706
mcUseConflict	-0.405	0.145	-0.698	-0.164

How account for dependencies through the network

Intuitively¹, we would want the response of i and j **not** to be independent

$$\Pr(Y_i = 1, Y_j = 1 \mid Y_{-ij}) \neq \Pr(Y_i = 1 \mid Y_{-ij}) \Pr(Y_j = 1 \mid Y_{-ij})$$

If there is a tie from i to j , $x_{ij} = 1$.

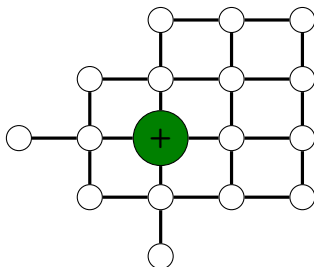
Suggesting a statistic

$$\sum_{i=1}^n \underbrace{y_i}_{\text{your succes}} \underbrace{\sum_{j \neq i} y_j x_{ij}}_{\text{\#successful friends}}$$

¹And this is what Robins et al., 2001, did

Ising model (Besag, 1972)

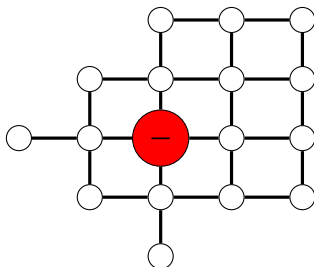
Probability spin $+$ $\approx$ # neighbours $j \in N(i)$ with spin $+$



$$\Pr(Y_i = 1 | Y_{N(i)} = y_{N(i)}) \propto \exp\{\theta_1 + \theta_2 \sum_{j \in N(i)} y_j\}$$

Ising model (Besag, 1972)

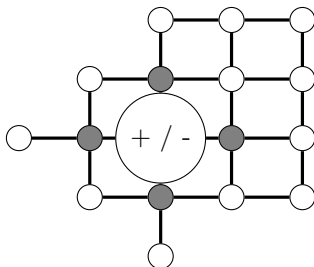
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Ising model (Besag, 1972)

Probability spin $+$ $\approx$ # neighbours $j \in N(i)$ with spin $+$



$$\Pr(Y_i = 1 | Y_{N(i)} = y_{N(i)}) \propto \exp\{\theta_1 + \theta_2 \sum_{j \in N(i)} y_j\}$$

Markov random fields for Social Networks

- ppl's' networks are not regular lattices
- ppl's' attitudes/behaviours also depend on SES, SEX, Education, etc



Social dependence is messy

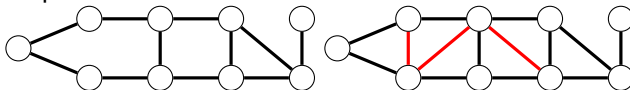
In Graphical models

Conditional independence graph: $i \sim j$ unless

$$X_i \perp X_j | X_{V \setminus \{i,j\}}$$

each node represents one variable (with many observations)

some dependence structures are easier than others



not decomposable

decomposable

Adding dependence between outcomes

Network Block Attribute Block

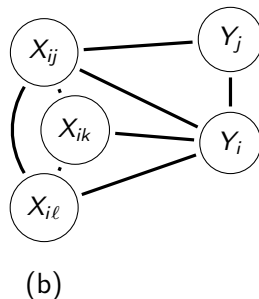
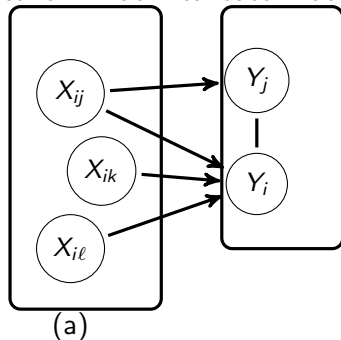


Figure: Dependence graph (a) and Moral graph (b) of model with dependence between attributes that share tie-variables

Deriving contagion statistics is non-trivial

To derive a non-trivial set of statistics use *realization-dependence* (Baddeley & Möller, 1989).

- Partial dependence graph $\mathcal{Q}_{\mathcal{B}}$, is a graph on $\mathcal{V}_{-\mathcal{B}}$
- where $\{i, j\} \in \mathcal{Q}_{\mathcal{B}}$ if
 - ✓ variables i and j are not conditionally independent conditional on variables $\mathcal{V}_{-\mathcal{B}, i, j}$,
 - ✓ and all variables corresponding to the index set $\mathcal{B}$ are zero.

In the model, the parameter for the statistic $A \subset \mathcal{V}$ is non-zero only if A is a clique of $\mathcal{M}$ and A is a clique of $\mathcal{Q}_{\mathcal{B}}$ for **all** $\mathcal{B}$.

Daraganova (2009) - derived statistics



Standard ALAAM

From this, and

- Making some Homogeneity assumptions and
- setting some higher-order statistics to zero,

we arrive at the following contagion model

$$p_{\theta}(\mathbf{y}|\mathbf{X}) = \exp \left\{ \theta_0 \sum_{i=1}^n y_i + \theta_{out} \sum_{i=1}^n y_i \sum_{j \neq i} x_{ij} + \theta_{in} \sum_{i=1}^n y_i \sum_{j \neq i} x_{ji} + \theta_{con} \sum_{i,j:i \neq j} y_i y_j (x_{ij} + x_{ji}) - \psi(\theta) \right\}$$

This includes an interaction term similar to that of Besag's (1972) classic auto-logistic model but it is subtly different in the definition of the neighbourhood.



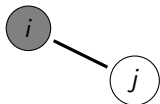
Auto-Logistic Actor Attribute Model (ALAAM)

ALAAM defines a distribution on **attributes** $\mathbf{y} \in \mathcal{Y} = \{0, 1\}^V$

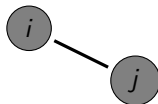
ALAAM pmf

$$p_{\theta}(\mathbf{y}|\mathbf{X}) = \exp\{\theta^{\top} z(\mathbf{y}; \mathbf{X}) - \psi(\theta)\}$$

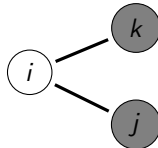
ERGM-like model for cross-sectional contagion, e.g.



$\sum_i y_i x_{i+}$ (degree)



$\sum_{i,j} y_i y_j x_{ij}$ (contagion)



$\sum_{i,j} y_i y_k x_{ij} x_{jk}$ (indirect contagion)

Example: ALAAM with social influence (1)

Example (Modern contraceptive use in rural Kenya contagion
 $n = 1303$)

Using balaam

```
res.DC.0 <- estimate.alaam( mcUse ~ odegree
                             + Age
                             + Female
                             + HasChildren
                             + relevanOthersApprove
                             + relevanOthersUse
                             + mcUseConflict
                             + simple,
                             village , adjacency = adj,
                             Iterations = 3000)
```

Example: ALAAM with social influence (2)

Example (MCU in rural Kenya contagion (cont.))

Using balaam

summaries of the posterior draws:

	mean	sd	ESS	SACF 10	SACF 30
intercept	-0.7396	0.2525	52.2484	0.6884	0.2903
contagion	0.4559	0.0826	62.0403	0.6853	0.3462
odegree	0.0282	0.0957	69.9401	0.6323	0.2418
Age	-0.0502	0.0071	56.9350	0.6882	0.3132
Female	-1.0827	0.1581	47.0392	0.7345	0.4571
HasChildren	1.7128	0.1959	59.1849	0.6733	0.3360
relevanOthersApprove	1.4891	0.1684	48.6652	0.7474	0.3920
relevanOthersUse	0.3036	0.1929	51.5122	0.6982	0.3561
mcUseConflict	-0.3744	0.1560	47.6004	0.7581	0.4128

if ESS is small you need to increase the number of Iterations

if and/or increase the parameter thinning which is currently 1 .

Alternatively, adjust the proposals by improving the proposal variance-co

Example: ALAAM with social influence (3)

Example (MCU in rural Kenya contagion (cont.))

Estimated ALAAM

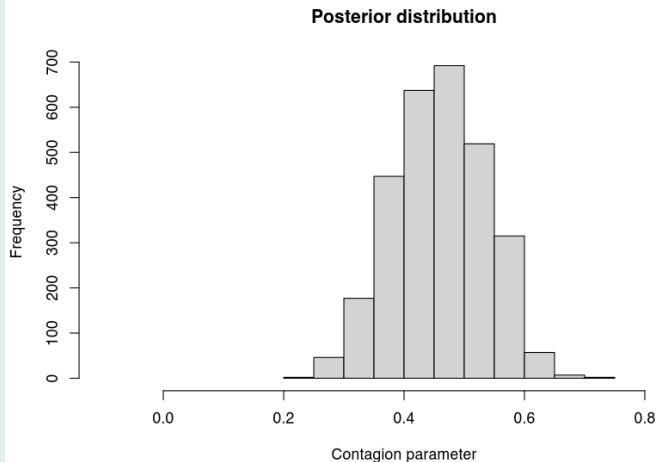
	Posterior		95% CI	
	Estimate	sd	0.025	0.975
intercept	-0.74	0.257	-1.283	-0.269
contagion	0.461	0.077	0.308	0.597
odegree	0.023	0.094	-0.182	0.195
Age	-0.05	0.007	-0.065	-0.036
Female	-1.082	0.16	-1.432	-0.77
HasChildren	1.709	0.198	1.303	2.101
relevanOthersApprove	1.489	0.167	1.172	1.781
relevanOthersUse	0.303	0.196	-0.096	0.68
mcUseConflict	-0.374	0.158	-0.668	-0.081

How much is the increase in the probability of mcUse if your friend uses?

Modi, Koskinen, DeChurch, Contractor, 2025, SocNet

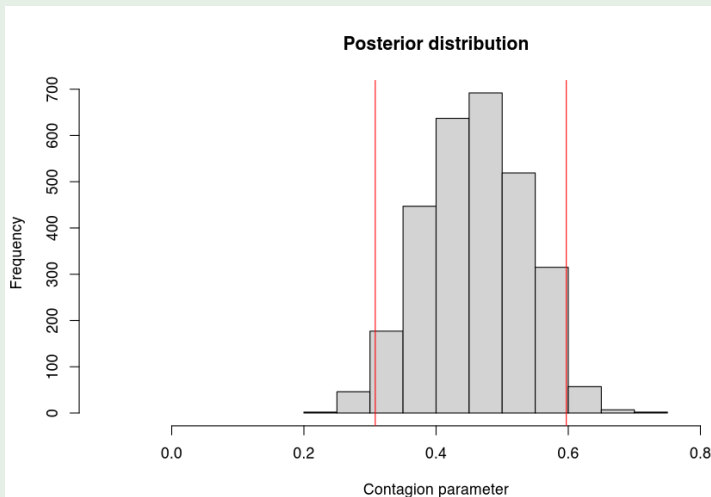
Example: ALAAM with social influence (4)

Example (MCU in rural Kenya contagion (cont.))



Example: ALAAM with social influence (4)

Example (MCU in rural Kenya contagion (cont.))



Effects: Covariates

Covariate effects are specified in standard `lm` syntax. For example, the term

$$\sum_{i=1}^n y_i w_{ij}$$

where the name of column j

For a model **dependent variable** agree,
and **covariates** mood and sex

$$\text{agree} \sim \text{odegree} + \text{mood} + \text{sex}$$

specifies the model defined by

$$z(\mathbf{y}, \mathbf{X}, \mathbf{W}) = \begin{bmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n y_i w_{i,\text{mood}} \\ \sum_{i=1}^n y_i w_{i,\text{sex}} \end{bmatrix}$$



Effects: Interactions of Covariate

To define and add an **interaction effect** between mood and sex,

`agree ~ odegree + mood + sex + mood * sex`

which would specify the model defined by

$$z(\mathbf{y}, \mathbf{X}, \mathbf{W}) = \begin{bmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n y_i w_{i,mood} \\ \sum_{i=1}^n y_i w_{i,sex} \\ \sum_{i=1}^n y_i w_{i,mood} w_{i,sex} \end{bmatrix}$$

Effects: Network activity effects (Markov)

Effect name	Formula u_i	Interpretation
degree	$y_i x_{i\cdot} = y_i \sum_j x_{ij}$	degree centrality and the probability of success
idegree	$y_i x_{\cdot i} = y_i \sum_j x_{ji}$	in-degree centrality and the probability of success
odegree	$y_i x_{i\cdot} = y_i \sum_j x_{ij}$	out-degree centrality and the probability of success
recipties	$y_i \sum_j x_{ij} x_{ji}$	strong ties and the probability of success
twostar	$y_i \binom{x_{i\cdot}}{2}$	centrality over and above degree
intwostar	$y_i \binom{x_{\cdot i}}{2}$	indegree centrality over and above indegree
outtwostar	$y_i \binom{x_{i\cdot}}{2}$	outdegree centrality over and above outdegree
threestars	$y_i \binom{x_{i\cdot}}{3}$	degree centrality over and above twostars
twopath	$y_i (x_{\cdot i} x_{i\cdot} - \sum_j x_{ij} x_{ji})$	association of brokerage on the probability of success.
inthreestar	$y_i \binom{x_{\cdot i}}{3}$	indegree centrality over and above intwostars
outthreestar	$y_i \binom{x_{i\cdot}}{3}$	outdegree centrality over and above outtwostars
transties	$y_i \sum_{j,k \neq i} x_{ij} x_{ik} x_{jk}$	embedded in (transitive) triads
indirties	$y_i \sum_j x_{ij} \sum_k (1 - x_{ik}) x_{jk}$	For (directed) undirected networks, the effect on probability of success of having ties to people that have ties to many people you are not directly tied to (see 3.1.3 Indirect network and contagion dependencies, Koskinen and Daraganova, for details)

Effects: Contagion effects

Effect name	Formula u_i	Interpretation
simple	$\sum_{i,j} y_i y_j x_{ij}$	is the probability of success increased by being connected to actors whose outcome is a success
recip	$\sum_{i,j} y_i y_j x_{ij} x_{ji}$	is the probability of success increased by being <i>mutually</i> tied to actors whose outcome is a success (directed networks only)
indirect	$\sum_i y_i \sum_j x_{ij} \sum_{k \neq i,j} y_k x_{jk}$	is the probability of success increased by being indirectly connected to actors whose outcome is a success (see 3.1.3 Indirect network and contagion dependencies, Koskinen and Daraganova, for details))
closedind	$\sum_i y_i \sum_j x_{ij} \sum_{k \neq i,j} y_k x_{ik} x_{jk}$	is the probability of success increased by being both indirectly and directly connected to actors whose outcome is a success (see 3.1.3 Indirect network and contagion dependencies as well as supplementary material, Koskinen and Daraganova, for details))
transitive	$\sum_i y_i \sum_j x_{ij} y_j \sum_{k \neq i,j} y_k x_{ik} x_{jk}$	is the probability of success increased by being embedded in triads where the two other members have success on the outcome(see 3.1.3 Indirect network and contagion dependencies as well as supplementary material, Koskinen and Daraganova, for details))

The network activity ALAAM - social influence

A closer look at the pmf

$$p(\mathbf{y} \mid \mathbf{X}) = \exp\{\theta^\top z(\mathbf{y}; \mathbf{X}) - \underbrace{\psi(\theta)}_{\text{norm. const.}}\} = \frac{e^{\theta^\top z(\mathbf{y}, \mathbf{X})}}{\underbrace{\sum_{\mathbf{y} \in \mathcal{X}} e^{\theta^\top z(\mathbf{y}, \mathbf{X})}}_{2^n \text{ terms}}}$$

We can **only** evaluate *conditional* probabilities

$$\Pr(Y_i = 1 \mid \mathbf{X}, \mathbf{y}_{-i}) = \frac{e^{\theta^\top z(\mathbf{y}^{i+}, \mathbf{X})}}{e^{\theta^\top z(\mathbf{y}^{i+}, \mathbf{X})} + e^{\theta^\top z(\mathbf{y}^{i-}, \mathbf{X})}}$$

where $\mathbf{y}^{i+}$ is $\mathbf{y}$ with $y_i = 1$, and $\mathbf{y}^{i-}$ is $\mathbf{y}$ with $y_i = 0$



Estimation

Markov chain Monte Carlo



Example: ALAAM with social influence - posteriors

Example (MCU in rural Kenya contagion (cont.))

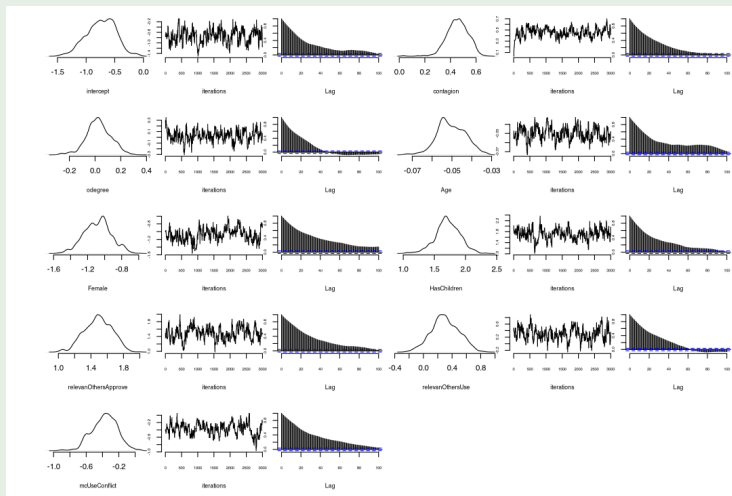
The ANOVA table of estimation results is a summary of *posteriors*

The algorithm **simulates** from these posteriors

```
plotPost(res.DC.0,figname='Kenya DC',showplot=TRUE)
```

Example: ALAAM with social influence - posteriors

Example (MCU in rural Kenya contagion (cont.))



Simulating from likelihood

We cannot evaluate likelihood for any θ , but for any θ we can simulate Y_i given $y_1, \dots, y_{i-1}, y_{i+1}, \dots, y_n$ using probabilities

$$\text{logit} \left\{ \Pr_{\theta}(Y_i = 1 | \mathbf{y}_{-i}, \mathbf{X}) \right\} = \theta^{\top} \{ z(\mathbf{y}^{i+}, \mathbf{X}) - z(\mathbf{y}^{i-}, \mathbf{X}) \}$$

giving us samples from

$$\mathbf{y} \mid \mathbf{X}, \theta$$

We will use this for

- estimation, and
- goodness-of-fit (GOF)

MPNet uses samples in stochastic approximation for MLE



Simulating from likelihood: Metropolis algorithm

Initialising in vector $\mathbf{y} := \mathbf{y}_0$, in each iteration t

- ① Pick $i \in V$ at random
- ② Propose to set $y_i := 1 - y_i$
- ③ Accept and set $\mathbf{y}_t := \Delta_i \mathbf{y}$, with probability

$$\min \left\{ 1, \exp\{\theta^\top [z(\Delta_i \mathbf{y}, \mathbf{X}) - z(\mathbf{y}, \mathbf{X})]\} \right\}$$

- ④ Otherwise set $\mathbf{y}_t := \mathbf{y}_{t-1}$

This gives us a sequence

$$\underbrace{\mathbf{y}_0, \mathbf{y}_1, \dots, \mathbf{y}_k}_{\text{first } k \text{ will remember } \mathbf{y}_0}, \overbrace{\mathbf{y}_{k+1}, \dots, \mathbf{y}_{T+1}, \mathbf{y}_T}^{\text{a dependent sample}}$$

For sufficiently large burnin k , $\mathbf{y}_{k+1}$ a draw from model.



MCMC for un-normalized distributions

MCMC: Sample $\theta^{(0)}, \theta^{(1)}, \dots$ from $\pi(\theta)$ by

- propose update $\theta^{(t)}$ to θ^* $q(\theta^*|\theta^{(t)})$
- set $\theta^{(t+1)} := \theta^*$ w.p. $\min\{1, H\}$

$$H = \frac{\pi(\theta^*)}{\pi(\theta^{(t)})} \frac{q(\theta^{(t)}|\theta^*)}{q(\theta^*|\theta^{(t)})}$$

(Works when $\pi(\theta) = f(\theta)/c(\theta)$ and $c(\theta)$ intractable)



Inference: ALAAM

For our target distribution $\pi(\theta|z)$

$$H = \frac{\exp\{\theta^{*\top} z(\mathbf{y}; \mathbf{X}) - \psi(\theta^*)\} \pi(\theta^*)}{\exp\{\theta^{(t)\top} z(\mathbf{y}; \mathbf{X}) - \psi(\theta^{(t)})\} \pi(\theta^{(t)})} \frac{q(\theta^{(t)}|\theta^*)}{q(\theta^*|\theta^{(t)})}$$

normalising constant $\psi(\cdot)$ of *likelihood* cannot be evaluated
(*model is doubly intractable*)



Solution to double intractability

Approximate $\hat{\lambda}(\theta, \theta^*) \approx \exp\{\psi(\theta) - \psi(\theta^*)\}$

- off-line importance sample (Koskinen, 2004)
- 'exact' auxiliary variable-based online importance sample with sample size of 1 - (Møller et al., 2006)
- 'exact' online (linked) path sampler auxiliary variable (Koskinen, 2008; Koskinen, 2009)
- online self-tuning auxiliary variable (Murray et al., 2006)
[Approximate Exchange Algorithm]

ERGO: we can obtain posterior for θ when y is observed



Monitoring performance of MCMC

Ideally, in our MCMC sample

$$\theta^{(0)}, \theta^{(1)}, \dots, \theta^{(M)}$$

the samples points are independent draws

$$\theta^{(m)} \stackrel{iid}{\sim} \pi(\theta|\mathbf{y}, \mathbf{X})$$

so that we use Monte Carlo estimators

$$\hat{\mathbb{E}}(\theta|\mathbf{y}, \mathbf{X}) = \bar{\theta} = \frac{1}{M} \sum_{m=1}^M \theta^{(m)}, \text{ and } \widehat{\mathbb{Cov}}(\theta|\mathbf{y}, \mathbf{X}) = \frac{1}{M} \sum_{m=1}^M (\theta^{(m)} - \bar{\theta})(\theta^{(m)} - \bar{\theta})^\top$$

as well as approximate probabilities $\Pr(\theta \in C)$, for any $C \subset \Theta$



Monitoring performance of MCMC - trace plots

In plots, *trace plots*, of

$$\theta^{(0)}, \theta^{(1)}, \dots, \theta^{(M)}$$

we should **not** see any

- trend/drift (independence of starting point)
 - ▶ select the number of initial iterations to discard - burnin
- serial correlation (good mixing)
 - ▶ space out sample points $\theta^{(k)}, \theta^{(2k)}, \theta^{(3k)}, \dots$ - thinning of sample



Monitoring performance of MCMC - SACF & ESS

The *sample autocorrelation function* (**SACF**) measures serial correlation between sample points

$$\theta^{(m-k)}, \theta^{(m)}$$

at different lags k

If SACF at lag k is low, say 30 (SIC?), then taking every k 'th sample point will yield an approximately independent sample

The *effective sample size* (**ESS**) tells us roughly how many independent sample points we have

Improving mixing

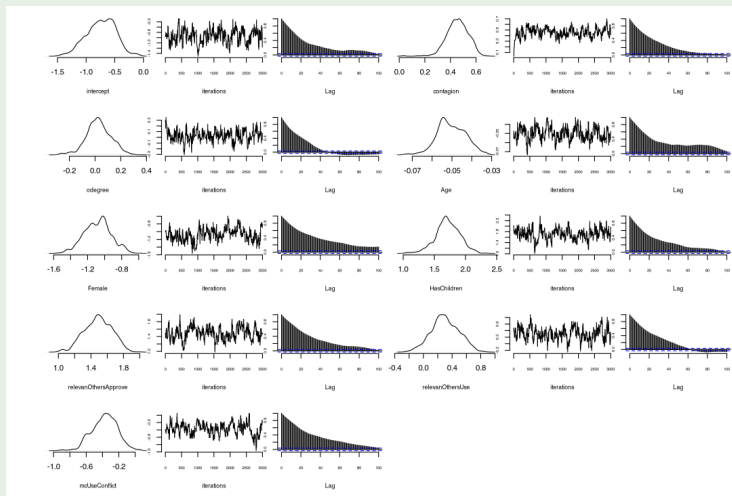
In our implementation the proposal distribution in each iteration

$$\theta^* \mid \theta^{(t)} \sim \mathcal{N}_p(\theta^{(t)}, \Sigma_p)$$

SACF can be lowered and mixing improved through improved Σ_p .

Example: ALAAM with social influence - posteriors

Example (MCU in rural Kenya contagion - convergence?)



Example: ALAAM with social influence - posteriors (2)

Example (MCU in rural Kenya contagion - convergence?)

In output from balaam

summaries of the posterior draws:

	mean	sd	ESS	SACF 10	SACF 30
intercept	-0.7396	0.2525	52.2484	0.6884	0.2903
contagion	0.4559	0.0826	62.0403	0.6853	0.3462
odegree	0.0282	0.0957	69.9401	0.6323	0.2418
Age	-0.0502	0.0071	56.9350	0.6882	0.3132
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relevanOthersUse	0.3036	0.1929	51.5122	0.6982	0.3561
mcUseConflict	-0.3744	0.1560	47.6004	0.7581	0.4128

if ESS is small you need to increase the number of Iterations

if and/or increase the parameter thinning which is currently 1 .

Alternatively, adjust the proposals by improving the proposal variance-co

Goodness-of-fit

Goodness-of-fit



Goodness-of-fit (GOF)

Once we have a draw

$$\theta^{(0)}, \theta^{(1)}, \dots, \theta^{(M)}$$

from $\pi(\theta|\mathbf{y})$, we can generate draws

$$\mathbf{y}^{(0)}, \mathbf{y}^{(1)}, \dots, \mathbf{y}^{(M)}$$

each from

$$p_{\theta^{(m)}}(\mathbf{y}^{(m)}|\mathbf{X})$$

GOF evaluation

If

$$\mathbf{y}^{(0)}, \mathbf{y}^{(1)}, \dots, \mathbf{y}^{(M)}$$

are 'similar' to $\mathbf{y}$, then model has good fit

Further topics

Further complications



Further topics: topics

- What model is 'best'?
- Missing outcomes and missing network ties
- Improving MCMC performance
- Hierarchical ALAAM
- Multivariate ALAAM

